PCV VALVE (Continued)

routed into the combustion chamber based on intake manifold vacuum.

TYPICAL PCV valves are shown in (Fig. 5), (Fig. 6) and (Fig. 7).

When the engine is not operating, or during an engine pop-back, the spring forces the plunger back against the seat (Fig. 5). This will prevent vapors from flowing through the valve.

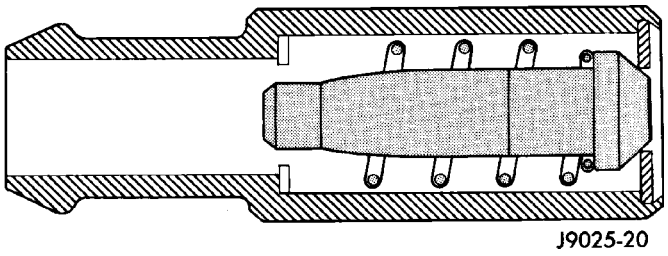


Fig. 5 ENGINE OFF OR ENGINE BACKFIRE - NO VAPOR FLOW

During periods of high manifold vacuum, such as idle or cruising speeds, vacuum is sufficient to completely compress spring. It will then pull the plunger to the top of the valve (Fig. 6). In this position there is minimal vapor flow through the valve.

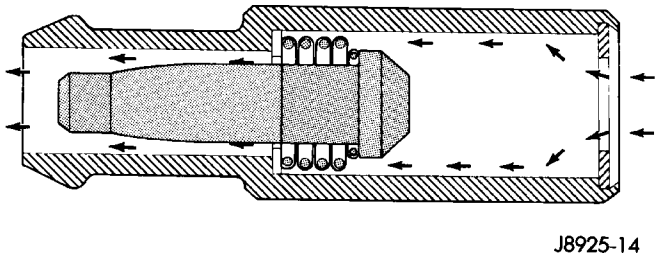


Fig. 6 HIGH INTAKE MANIFOLD VACUUM - MINIMAL VAPOR FLOW

During periods of moderate manifold vacuum, the plunger is only pulled part way back from inlet. This results in maximum vapor flow through the valve (Fig. 7).

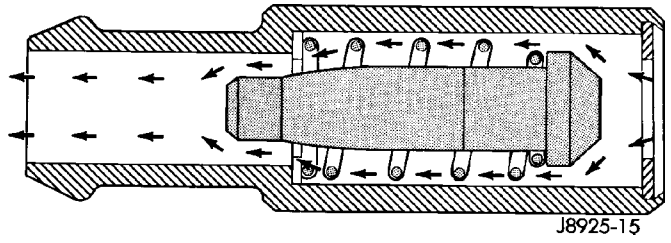


Fig. 7 MODERATE INTAKE MANIFOLD VACUUM - MAXIMUM VAPOR FLOW

REMOVAL

2.4L

The PCV valve is attached to the left/front side of the valve cover (Fig. 2). It is secured with 1 bolt. An o-ring is used to seal valve to valve cover (Fig. 3). Another type of threaded PCV valve may be used (Fig. 4).

(1) Remove hose from valve. Check condition of hose.

(2) Bolted PCV Valve:
  (a) Remove 1 bolt.
  (b) Remove PCV valve from valve cover.
  (c) Check condition of valve o-ring.

(3) Threaded PCV Valve:
  (a) Unscrew valve from valve cover.

EVAP/PURGE SOLENOID

DESCRIPTION

The duty cycle EVAP canister purge solenoid (DCP) regulates the rate of vapor flow from the EVAP canister to the intake manifold. The Powertrain Control Module (PCM) operates the solenoid.

OPERATION

During the cold start warm-up period and the hot start time delay, the PCM does not energize the solenoid. When de-energized, no vapors are purged. The PCM de-energizes the solenoid during open loop operation.

The engine enters closed loop operation after it reaches a specified temperature and the time delay ends. During closed loop operation, the PCM cycles (energizes and de-energizes) the solenoid 5 or 10 times per second, depending upon operating conditions. The PCM varies the vapor flow rate by changing solenoid pulse width. Pulse width is the amount of time that the solenoid is energized. The PCM adjusts solenoid pulse width based on engine operating condition.

REMOVAL

The duty cycle EVAP canister purge solenoid is located in the engine compartment near the front of the brake master cylinder (Fig. 8). It is attached to a slip-on type bracket.

The EVAP System Test Port is located near the EVAP/Purge Solenoid (Fig. 8).

(1) Disconnect electrical wiring connector at solenoid.

(2) Disconnect vacuum lines/hoses at solenoid.

(3) Lift solenoid and rubber solenoid support from mounting bracket.